

AGORA

Post-Quantum Security Audit Pack

Program: /home/king_d/solana/testing/marginfi

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Tool: Pinpoint SPECTRE 1.2.2

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Executive Summary

Total findings: 173

Cryptographic components inventoried (CBOM 1.6): 129

Findings by rule: CA-001=0, CA-002=100, CA-003=0, CA-004=0

CA-005: not applicable. PDAs are quantum-safe by construction.

Findings

CA-002

programs/marginfi/src/instructions/drift/add_pool.rs:117 | sev=Medium | Single-signer au
programs/marginfi/src/instructions/drift/deposit.rs:158 | sev=Medium | Single-signer au
programs/marginfi/src/instructions/drift/withdraw.rs:335 | sev=Medium | Single-signer au
programs/marginfi/src/instructions/juplend/add_pool.rs:124 | sev=Medium | Single-signer
programs/marginfi/src/instructions/juplend/deposit.rs:157 | sev=Medium | Single-signer a
programs/marginfi/src/instructions/juplend/withdraw.rs:351 | sev=Medium | Single-signer
programs/marginfi/src/instructions/kamino/add_pool.rs:111 | sev=Medium | Single-signer a
programs/marginfi/src/instructions/kamino/deposit.rs:159 | sev=Medium | Single-signer au
programs/marginfi/src/instructions/kamino/withdraw.rs:298 | sev=Medium | Single-signer a
programs/marginfi/src/instructions/marginfi_account/borrow.rs:256 | sev=Medium | Single-
programs/marginfi/src/instructions/marginfi_account/close.rs:31 | sev=Medium | Single-si
programs/marginfi/src/instructions/marginfi_account/close_balance.rs:77 | sev=Medium | S
programs/marginfi/src/instructions/marginfi_account/deposit.rs:168 | sev=Medium | Single
programs/marginfi/src/instructions/marginfi_account/emissions.rs:37 | sev=Medium | Singl
programs/marginfi/src/instructions/marginfi_account/flashloan.rs:54 | sev=Medium | Singl
programs/marginfi/src/instructions/marginfi_account/flashloan.rs:144 | sev=Medium | Sing
programs/marginfi/src/instructions/marginfi_account/freeze.rs:57 | sev=Medium | Single-s
programs/marginfi/src/instructions/marginfi_account/initialize.rs:53 | sev=Medium | Sing
programs/marginfi/src/instructions/marginfi_account/initialize.rs:122 | sev=Medium | Sin
programs/marginfi/src/instructions/marginfi_account/liquidate.rs:521 | sev=Medium | Sing
programs/marginfi/src/instructions/marginfi_account/liquidate_end.rs:285 | sev=Medium |
programs/marginfi/src/instructions/marginfi_account/liquidate_start.rs:274 | sev=Medium
programs/marginfi/src/instructions/marginfi_account/order.rs:537 | sev=Medium | Single-s
programs/marginfi/src/instructions/marginfi_account/order.rs:589 | sev=Medium | Single-s
programs/marginfi/src/instructions/marginfi_account/order.rs:632 | sev=Medium | Single-s
programs/marginfi/src/instructions/marginfi_account/purge_delev_balance.rs:66 | sev=Medi
programs/marginfi/src/instructions/marginfi_account/repay.rs:199 | sev=Medium | Single-s
programs/marginfi/src/instructions/marginfi_account/transfer_account.rs:116 | sev=Medium
programs/marginfi/src/instructions/marginfi_account/transfer_account.rs:260 | sev=Medium
programs/marginfi/src/instructions/marginfi_account/withdraw.rs:279 | sev=Medium | Singl
programs/marginfi/src/instructions/marginfi_group/add_pool.rs:106 | sev=Medium | Single-
programs/marginfi/src/instructions/marginfi_group/add_pool_with_seed.rs:108 | sev=Medium
programs/marginfi/src/instructions/marginfi_group/clone_bank.rs:155 | sev=Medium | Singl
programs/marginfi/src/instructions/marginfi_group/close_bank.rs:66 | sev=Medium | Single
programs/marginfi/src/instructions/marginfi_group/collect_bank_fees.rs:302 | sev=Medium
programs/marginfi/src/instructions/marginfi_group/collect_bank_fees.rs:381 | sev=Medium
programs/marginfi/src/instructions/marginfi_group/collect_bank_fees.rs:441 | sev=Medium
programs/marginfi/src/instructions/marginfi_group/config_bank_emode.rs:59 | sev=Medium |
programs/marginfi/src/instructions/marginfi_group/config_bank_oracle.rs:59 | sev=Medium
programs/marginfi/src/instructions/marginfi_group/config_group_fee.rs:14 | sev=Medium |

NIST IR 8547 Traceability Matrix

Maps each audit-pack section to a NIST IR 8547 post-quantum transition phase.

Phase	Rule(s)	Finding Count
Inventory	CA-001	0
Risk Prioritisation	CA-002, CA-003, CA-004	100
Hybrid Deployment	(no rules at this release)	n/a
Full Migration	(no rules at this release)	n/a

Hybrid Deployment and Full Migration are documented for framework completeness.

No CA rule fires for those phases at this release. Coverage expands in Tier 2.

Methodology

Pre-pass: Anchor expand subprocess invoked before tree-sitter walks source.

Expanded output is cached per program hash to avoid redundant re-expansion.

OtterSec downstream framing: CA-003 cross-program assumption findings surface migration cascade complexity following OtterSec's CPI-trust analysis framework.

CA-005 dropped: PDAs are quantum-safe by construction. Off-curve points derived via SHA-256 have no Ed25519 private key, so the Grover speedup falls beyond the attack horizon. Confirmed with Jump Crypto and Helius.

CA-004 is inventory and exposure only. It does not prescribe a hybrid Falcon co-signer. Protocol-layer migration (address format, ZK proof of Ed25519 seed) is owned by Anza and Firedancer, not by individual program developers.

Rule pack: pinpoint-rules-pq v0.1.0. CBOM format: CycloneDX 1.6.

PDF renderer: printpdf (pure Rust, no system library dependency).